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MASTER OF TECHNOLOGY IN SOFTWARE ENGINEERING

Graduate Certificate Examination
Sample Examination Paper

Subject: *Architecting Scalable Systems*

APPENDIX A

"Vote Anywhere"
Case Study

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(The content of this case study is fictitious!)

Around 5th century BCE, ancient Athens in ancient Greece were conducting voting among eligible citizens for various communal issues like electing officials, passing laws, and making policy decisions. The practice continues till today in many countries and regions especially for electing government leaders, representatives, etc.

While there were several instances of small-scaled voting being conducted using electronic systems, most of the instances of voting are still conducted using paper ballots and tallied manually despite many inconveniences like the following:

- Need to travel to potentially faraway and hard-to-access polling stations.
- Potentially long waiting time during peak voting periods.
- Potential inaccuracies and delays in counting and tabulation of paper ballots.
- Potentially long wait for the finalised results.

Electronic voting (i.e., e-voting) is ideal and desirable but faces a variety of challenges against its widespread adoption; the following are some of the challenges:

- Scepticism on the security of the systems against various security threats.
- Mistrust of the public on the integrity of the systems.
- Difficulties in developing robust systems and deploying them to meet the expected reliability, capacity, timeliness, etc.

You Vote is a local start-up whose founders are passionate and optimistic with the adoption of e-voting in smaller countries or regions having good digital infrastructure. The company has already planned the product roadmap for its e-voting platform named ***Vote Anywhere***. Being a small island nation with highly developed digital infrastructure, Singapore naturally becomes the first strategic testbed for the MVP of ***Vote Anywhere***.

Voting Anywhere would be developed over multiple phases. At present, only the initial two phases are planned. The first phase targets to implement the MVP that includes the features that provides some essential functions and a few select differentiated functions (e.g., near real-time update of results) for off-site virtual e-voting. The second phase targets to add more select differentiated functions (e.g., prediction of results).

You are the Lead Solution Architect assigned to the project. You are responsible for the technical delivery of the platform that includes the architecture, design, development and deployment of the platform and three pilot applications in the first phase. After the first phase completes successfully, the platform should allow for expansion of new services and inclusion of new actors. There are 6 pilot applications: 1 mobile application for the voters; and 5 web applications for the election admins, proctors, observers, public users and platform admins respectively.

1. Phase 1 of the Platform

The new platform supports the following actors:

1. Public User: Anyone may access the platform via its web frontend. He/she would normally register him/herself with the platform. Voters are an exception as they are pre-registered.
2. Platform User: Anyone, except for voters, may register as a user of the platform. An authenticated Platform User may act as an Election Admin, Proctor, Observer or Platform Admin. A new registration is to be reviewed and approved by a Platform Admin.
3. Election Admin: A Platform User who manages the polls to be conducted on the platform.
4. Proctor: A Platform User who verifies authenticated Voters via live video sessions.
5. Observer: A Platform User who are granted access to more detailed results and the poll report. He/she could be a candidate of the poll or a representative from a political party.
6. Voter: A Platform User who is pre-registered for a poll and would cast his/her vote via the platform.
7. Platform Admin: Member of the platform provider who administers the Platform Users.
8. SingPass Authenticator: The external system of SingPass that serves as an OpenID Connect provider. It helps the platform to verify a Platform User via his/her SingPass credentials.
9. Email Gateway: The external third party system for sending emails to the users.

Figure 1 below illustrates the high-level functionality of Phase 1.

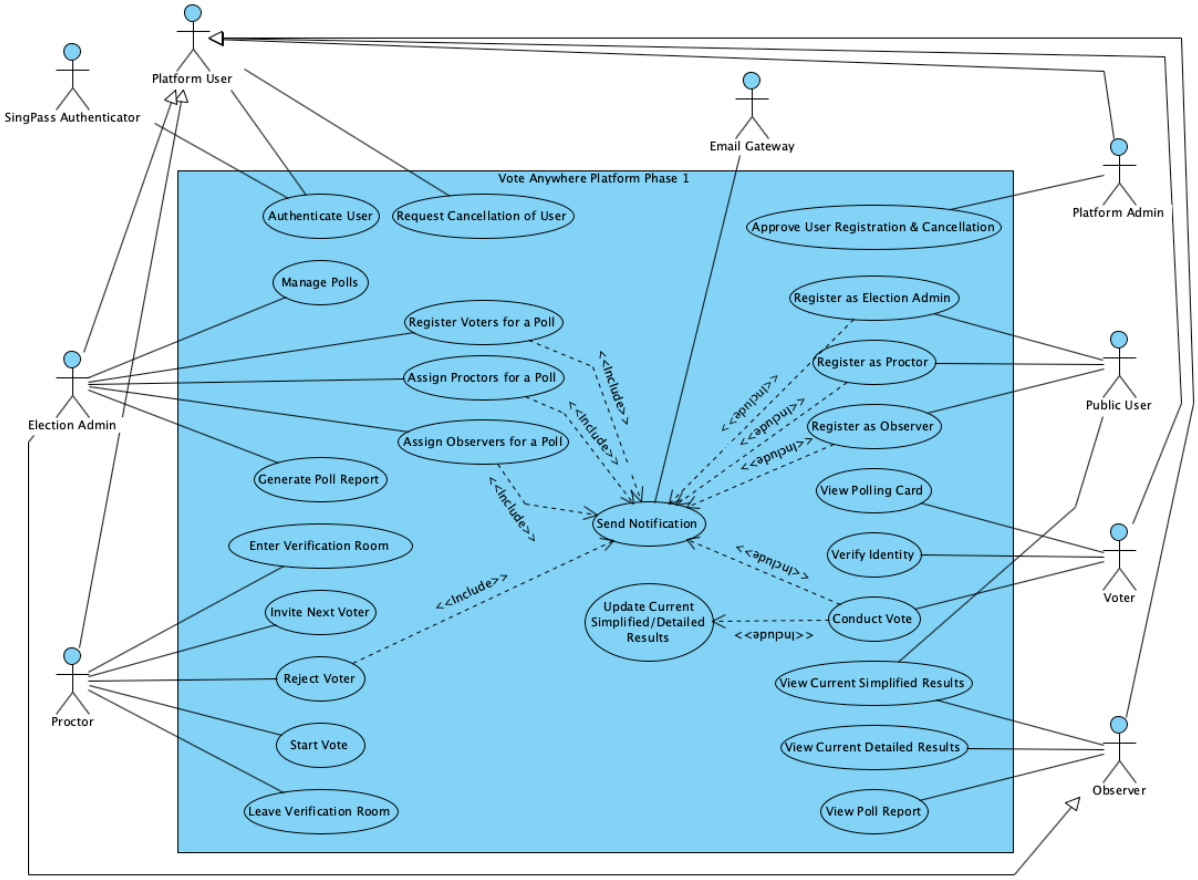


Figure 1: System Context for Phase 1

Here is a brief description of the use cases for Phase 1:

Use Case	Description
Register as Election Admin	The Public User registers him/herself with the system as an Election Admin by providing his/her particulars. The platform verifies his/her identity by sending a verification email.
Register as Proctor	The Public User registers him/herself with the system as a Proctor by providing his/her particulars. The platform verifies his/her identity by sending a verification email.
Register as Observer	The Public User registers him/herself with the system as an Observer by providing his/her particulars. The platform verifies his/her identity by sending a verification email.
Authenticate User	The Platform User logs in to the platform to start to use the platform features that require authentication.
Request Cancellation of User	The Platform User requests so that his/her account is removed from the platform. This does not apply to Voters.
Approve User Registration & Cancellation	The Platform Admin reviews the request by a Public User to register as a Platform User. The Platform Admin reviews the request by a Platform User to be removed from the platform. The system create/remove the user account accordingly.

Manage Polls	The Election Admin manages the polls that are administered by him/her. A poll includes details like poll type, date/time range, detailed description, eligible Voters, assigned Proctors, assigned Observers, etc. At present, only one kind of poll is supported: 1. Presidential Election. This is the simplest kind of poll where the president of the country is elected among two or more individual candidates. Each eligible Voter is entitled to cast one vote for any one of the candidates. The candidate with the highest number of votes would be the winner.
Register Voters for a Poll	The Election Admin registers all the eligible Voters by importing them in bulk from a file provided by the election commission of the country/region. The platform notifies the registered Voters by sending every Voter a polling notification email.
Assign Proctors for a Poll	The Election Admin assigns one or more registered Proctors to verify the Voters of a poll. The assignment includes the "polling room" as well. The platform notifies the assigned Proctors by sending every Proctor a proctoring notification email.
Assign Observers for a Poll	The Election Admin assigns one or more registered Observers to monitor the results of a poll. The platform notifies the assigned Observers by sending every Observer an observation notification email.
Generate Poll Report	The Election Admin generates the poll report after the conclusion of a poll. The poll report is the formal deliverable of a poll; it contains all the details of the poll as well as the simplified and detailed results.
View Polling Card	The Voter views his/her polling card containing the details of the upcoming poll as well as his/her registered particulars.
Verify Identity	During the valid duration of the poll, the Voter starts his/her voting process by selecting to verify his/her identity. He/she would be put on a virtual queue, waiting to be verified by one of the assigned Proctors. Upon his/her turn for verification, he/she would be admitted into the "polling room" where he/she would be in a video conference session with a Proctor; he/she would have to answer a few questions verbally to verify his/her identity. He/she would then be automatically transitioned to the *Conduct Vote* use case.
Conduct Vote	While in the "polling room", the Voter selects his/her choice of candidate from the list of candidates for the poll. Upon the confirmation of his/her choice, the Voter is removed from the "polling room" by the Proctor. The platform notifies the Voter by sending a voting completion notification email. The platform would also update the current simplified/detailed results by invoking the *Update Current Simplified/Detailed Results* use case. Note that this use case also captures the details of voting, excluding the choice of candidate, and stores them locally on the Voter device. The locally stored details can be used in future as a proof of voting where relevant.
Enter Verification Room	During the valid duration of the poll, the Proctor starts his/her proctoring session by selecting to enter his/her assigned "verification

	room". He/she is admitted into the "polling room" where he/she would be in a video conference with Voters, one at a time.
Invite Next Voter	While in the "polling room", the Proctor selects to admit the next Voter to be verified whenever he/she is ready. Upon admitting a Voter, he/she would verify the Voter's identity by asking the latter a few verification questions verbally. If the Proctor is satisfied with the responses, he/she would select to allow the Voter to start voting by invoking the *Start Vote* use case. However, if the Proctor is not satisfied with the responses; he/she would select to reject the Voter by invoking the *Reject Voter* use case.
Reject Voter	The platform removes the current Voter from the "polling room" and sends a rejection notification email to the Voter.
Start Vote	The platform displays the list of candidates of the poll for the Voter to make his/her choice. Upon confirming his/her choice, the platform removes the Voter from the "polling room.
Leave Verification Room	During the valid duration of the poll, when not actively verifying a Voter, the Proctor may terminate his/her proctoring session by selecting to leave his/her assigned "polling room".
View Current Simplified Results	During the valid duration of the poll and a configurable period after the conclusion of the poll, the Public User or Observer may view the current simplified results of the poll. It contains the current tallied results including the counts and percentages of votes for each candidate. If the user stays in the same screen, the results would be automatically updated whenever there are new changes.
View Current Detailed Results	During the valid duration of the poll, the Observer may view the current detailed results of the poll. It contains the current tallied results (as presented as current simplified results) as well as other details including the breakdown of vote counts by voter constituencies, by voter age groups, etc.
View Poll Report	After the conclusion of the poll, the poll report is made available to the Observer for a configurable period. The Observer may view the poll report which contains all the details of the poll as well as the simplified and detailed results.
Send Notification	When an use case needs to send an email notification, this use case in invoked to send the email to the recipient(s) via the Email Gateway.

2. Phase 2 of the Platform

In addition to the Phase 1 functionality, this platform is being extended with features that cater to accessibility as well as prediction of the trends of results.

There are no new actors.

Figures 2 below illustrates the high-level functionality of Phase 2.

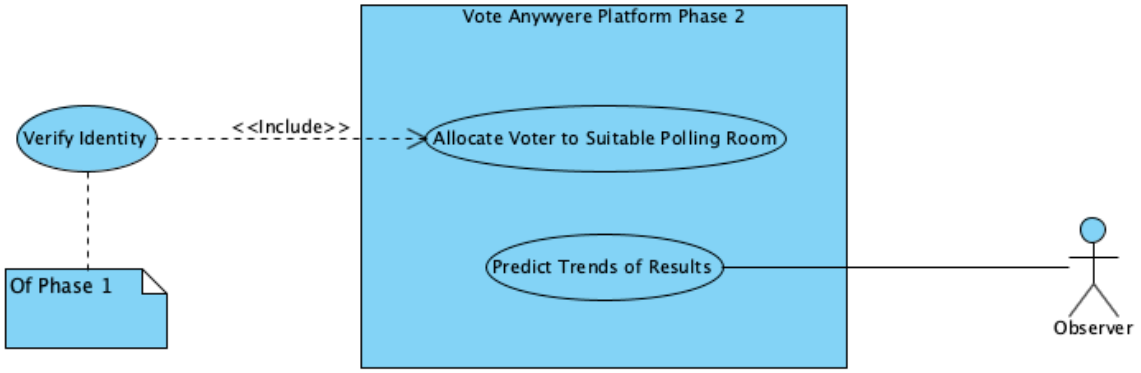


Figure 2: System Context for Phase 2

Here is a brief description of the use cases for Phase 2:

Use Case	Description
Allocate Voter to Suitable Polling Room	As some of the Voters may have specific special needs like the use of sign language, limited language options, etc., the platform should optimally allocate the Voters to the "verification room" that is manned by a Proctor of suitable profile. This use case is invoked whenever the Voter invokes the *Verify Identity* use case.
Predict Trends of Results	The platform allows the Observer to predict the trends of results based on some parameters that can be configured based on the preferences and assumptions of the Observer. These parameters include voter turnout (i.e., some voters may turn up later), voter demographic (i.e., voters of some demographic may turn up later), etc.

3. Extensibility of the Platform

The platform is designed to use common services and libraries that support and speed up the development of future services provided by the platform.

4. Performance and Scalability of the Platform

The platform is designed for the following performance and scalability:

- For phases 1 and 2, the platform supports up to 10 concurrent Election Admins; 1,000 concurrent Proctors; 1,000 concurrent Observers; 100,000 concurrent Voters and 5 concurrent Platform Admins.
- For interactive request-response type of operation (e.g., Verify Identity), the expected response time is less than 3 seconds for 90% of the requests.
- The platform is currently not expected to handle multitenancy (i.e., multiple countries or regions).
- The platform handles increasing amount of data while maintaining the response time of the application.

- The platform handles near real-time update of current simplified/detailed results. Update of current simplified results should be completed within 1 minute on average while update of current detailed results should be completed within 3 minutes on average.
- The platform should ensure isolation of data without sacrificing the consistency of the update and retrieval of data processed in real-time.

5. Security Architecture of the Platform

The platform is designed with the following security considerations:

- The system follows general security best practices in infrastructure and application security e.g., reverse proxy in DMZ, etc.
- Personal information is stored in encrypted form and the platform is able to find all information stored about an individual. According to GDPR, the actors should allow users to request access to user's data and the platform should be able to facilitate the compliance.

6. Initial Architecture of the Platform

As the Lead Solution Architect, you have provided the following architectural guidelines for the platform and the applications.

6.1 Architecture Layers

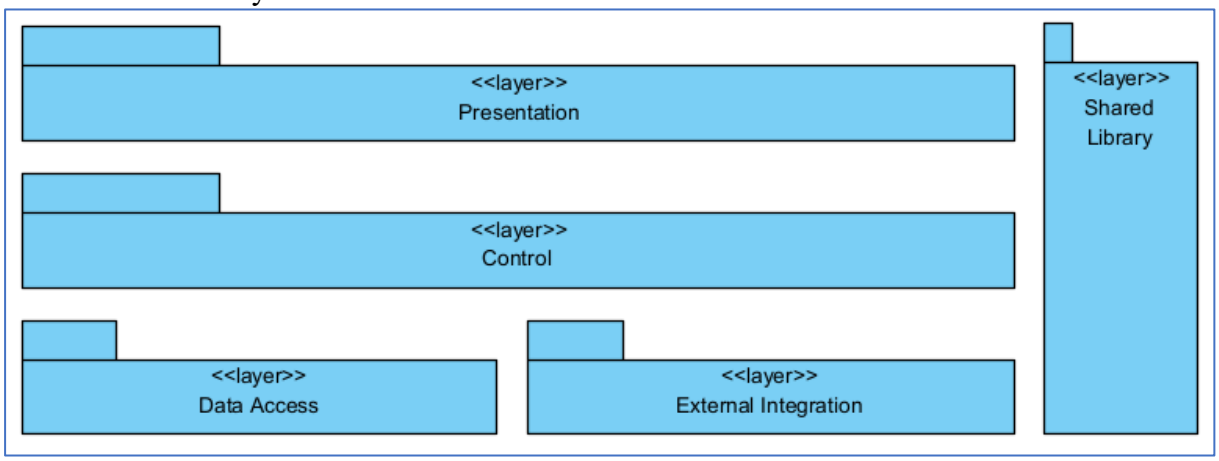


Figure 3: Architecture Layers

The above layered architecture illustrates the desirable layering of a typical application on the platform. The Presentation layer handles the user interaction of the application. The Control layer handles the flow logics of the application use cases. The Data Access layer handles the access to persistent storages. The External Integration layer handles the integration with external systems. The Shared Library layer hosts the shared/reusable libraries.

6.2 Logical Architecture Overview

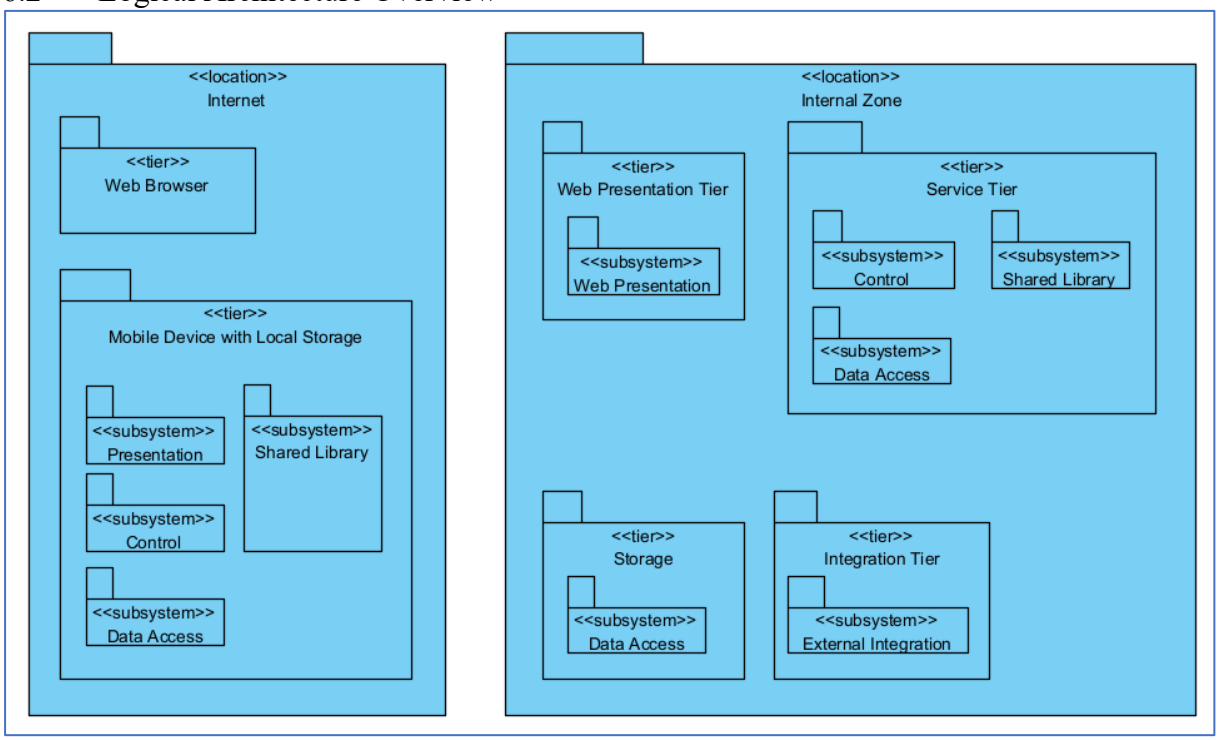


Figure 4: Architecture Overview

The above logical architecture overview illustrates the desirable tiering of the platform. The frontend tiers are in the form of Web Browser (for the Election Admins, Proctors, Observers, Public Users and Platform Admins), Mobile Device with Local Storage (for Voters). The backend tiers are Web Presentation, Service, Storage and Integration.

6.3 Deployment Options

The platform is to be designed as a cloud native solution. The services should be implemented as microservices.

7. Domain Model for Phase 1

A Software Architect has analyzed the requirements for Phase 1 and identified the following domain entities and services. The Development team would implement the platform according to this domain model.

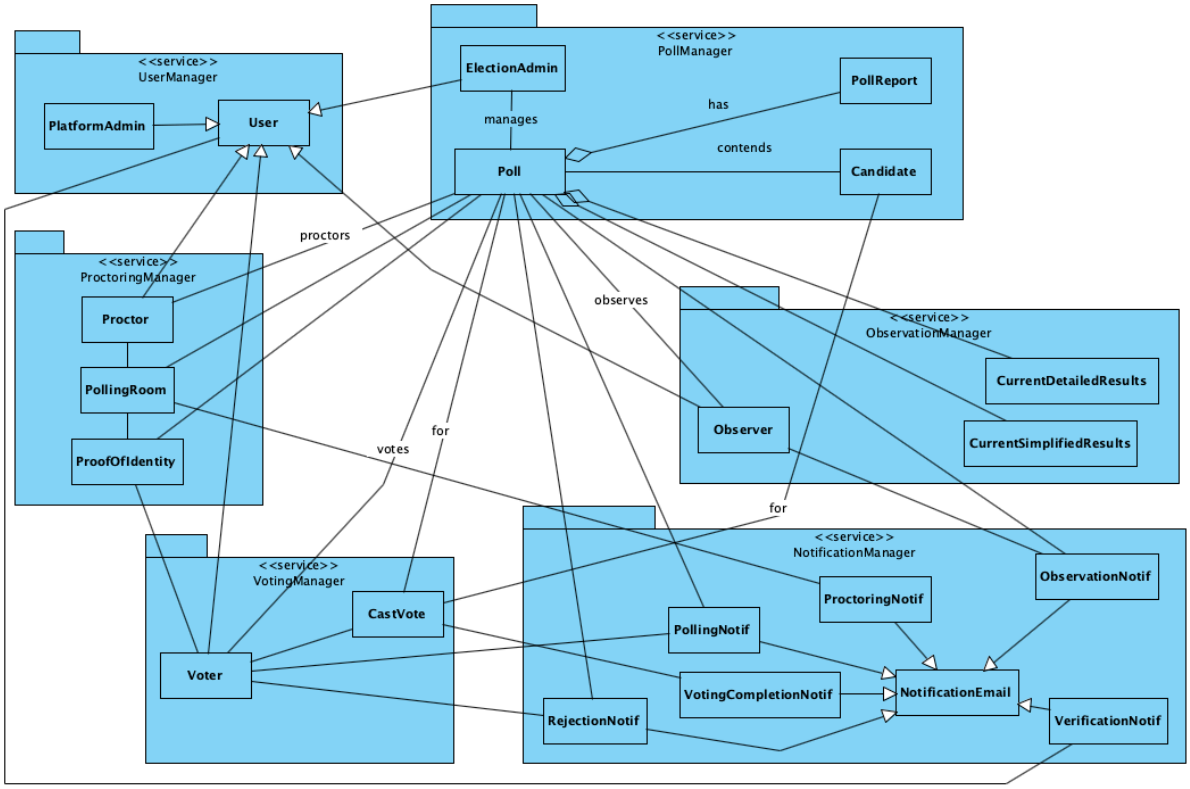


Figure 5: Domain Model for Phase 1

8. Logical Packaging cum Deployment Model for Phase 1

A Solution Architect has crafted an **incomplete** and **potentially erroneous** logical architecture for the *Conduct Vote* use case. The emphasis is on the packaging of the associated functional elements into subsystems as well as on the deployment of the subsystems to the mobile devices of Voters. Do take note that the diagram intentionally omits the deployment for the backend of the platform.

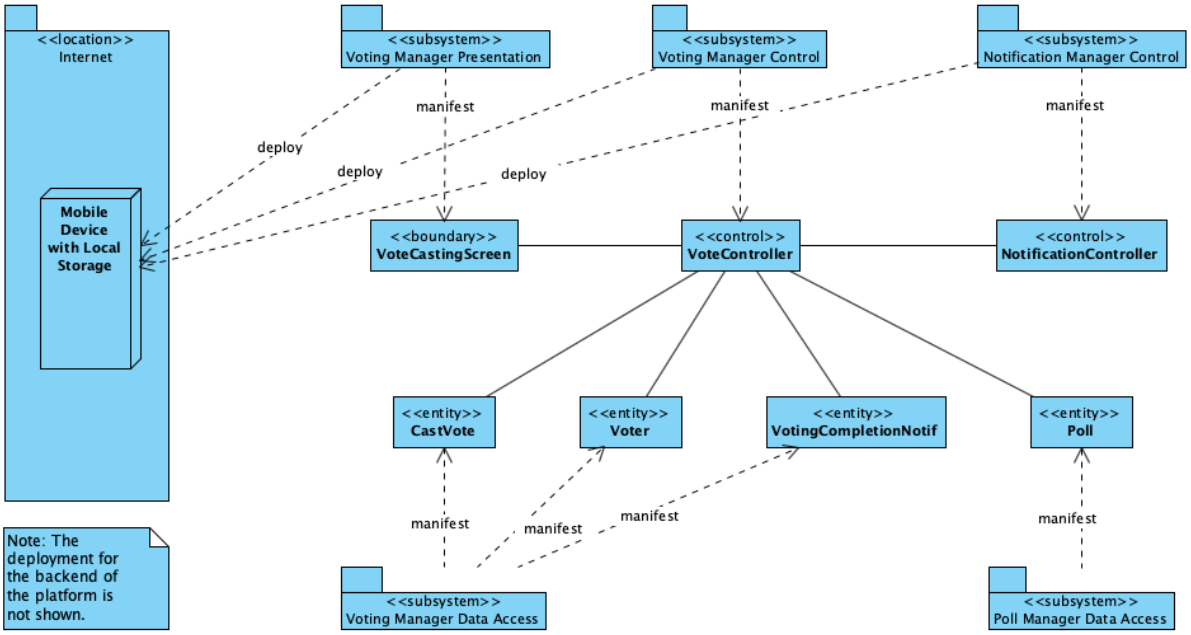


Figure 6: Logical Packaging cum Deployment Model for Phase 1

END OF CASE STUDY